Configuração 6: M = 128, C = 8, K = 8, IoT = 20 dB, QPSK
Perda de SNR do BCD-MMSE (4 iterações) em SER = 10^{-2}, em dB, em relação à referência ideal (float64); × = não chega a 10^{-2} nas SNRs simuladas
Fundo laranja no rótulo: o modo com o melhor resultado no par, comparando do formato com menos bits para o de mais, sem contar as colunas em que todos os modos ficam abaixo de 0,05 dB
Embaixo da perda: maior ocupação da XCU50 entre LUT, DSP, FF e BRAM (síntese HLS, sem diretivas), no mesmo formato e modo
* a referência já está abaixo de 10^{-2} no primeiro SNR simulado: a perda é um limite inferior

Perda de SNR (dB)

< 0,1 dB | 0,1 a 0,5 dB | 0,5 a 1 dB | 1 a 2 dB | 2 a 3 dB | 3 a 5 dB | 5 a 10 dB | > 10 dB | × não chega a 10^{-2} | rótulo do melhor modo

TDLA30-5 *

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	4.62 1.8%	1.06 2.1%	0.01 3.1%	0.00 3.6%
	AP_SAT_ZERO	4.62 1.8%	0.77 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	4.62 1.9%	1.06 2.2%	0.01 3.7%	0.00 4.2%
	AP_WRAP	4.62 1.7%	0.01 2.3%	0.01 3.5%	0.00 4.1%
	AP_WRAP_SM	4.62 1.8%	1.13 2.2%	0.01 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	× 2.0%	1.05 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	× 1.8%	0.76 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	× 2.2%	1.05 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	× 1.6%	0.01 2.0%	0.01 3.2%	0.00 3.7%
	AP_WRAP_SM	× 2.0%	1.11 2.4%	0.01 3.7%	0.00 4.2%
AP_RND	AP_SAT	2.62 2.0%	1.06 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.62 1.8%	0.77 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	2.62 2.1%	1.06 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	2.62 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	2.62 2.0%	1.13 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	2.62 2.0%	1.06 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.62 1.8%	0.77 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	2.62 2.2%	1.06 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	2.62 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	2.62 2.0%	1.13 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	2.63 1.9%	1.06 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.63 1.8%	0.77 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	2.63 2.1%	1.06 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	2.63 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	2.63 2.0%	1.13 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	2.62 2.0%	1.06 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.62 1.8%	0.77 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	2.62 2.2%	1.06 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	2.62 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	2.62 2.0%	1.13 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	2.62 2.0%	1.06 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.62 1.8%	0.77 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	2.62 2.2%	1.06 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	2.62 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	2.62 2.0%	1.13 2.4%	0.00 3.7%	0.00 4.2%

TDLA30-75

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	4.81 1.8%	1.17 2.1%	0.04 3.1%	0.00 3.6%
	AP_SAT_ZERO	4.81 1.8%	0.85 2.4%	0.04 3.7%	0.00 4.2%
	AP_SAT_SYM	4.81 1.9%	1.17 2.2%	0.04 3.7%	0.00 4.2%
	AP_WRAP	4.81 1.7%	0.05 2.3%	0.04 3.5%	0.00 4.1%
	AP_WRAP_SM	4.81 1.8%	1.19 2.2%	0.04 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	× 2.0%	1.15 2.3%	0.03 3.7%	0.00 4.2%
	AP_SAT_ZERO	× 1.8%	0.84 2.4%	0.03 3.7%	0.00 4.2%
	AP_SAT_SYM	× 2.2%	1.15 2.5%	0.03 3.7%	0.00 4.2%
	AP_WRAP	× 1.6%	0.04 2.0%	0.03 3.2%	0.00 3.7%
	AP_WRAP_SM	× 2.0%	1.17 2.4%	0.03 3.7%	0.00 4.2%
AP_RND	AP_SAT	2.67 2.0%	1.16 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.67 1.8%	0.85 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.67 2.1%	1.16 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.67 1.7%	0.02 2.3%	0.01 3.5%	0.00 4.1%
	AP_WRAP_SM	2.67 2.0%	1.18 2.3%	0.01 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	2.67 2.0%	1.16 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.67 1.8%	0.85 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.67 2.2%	1.16 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.67 1.6%	0.02 2.0%	0.01 3.2%	0.00 3.7%
	AP_WRAP_SM	2.67 2.0%	1.18 2.4%	0.01 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	2.67 1.9%	1.16 2.2%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.67 1.8%	0.85 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.67 2.1%	1.16 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.67 1.8%	0.02 2.4%	0.01 3.7%	0.00 4.2%
	AP_WRAP_SM	2.67 2.0%	1.18 2.3%	0.01 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	2.67 2.0%	1.16 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.67 1.8%	0.85 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.67 2.2%	1.16 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.67 1.6%	0.02 2.0%	0.01 3.2%	0.00 3.7%
	AP_WRAP_SM	2.67 2.0%	1.18 2.4%	0.01 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	2.67 2.0%	1.16 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.67 1.8%	0.85 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.67 2.2%	1.16 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.67 1.8%	0.02 2.4%	0.01 3.7%	0.00 4.2%
	AP_WRAP_SM	2.67 2.0%	1.18 2.4%	0.01 3.7%	0.00 4.2%

TDLA30-300

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	4.91 1.8%	1.17 2.1%	0.03 3.1%	0.00 3.6%
	AP_SAT_ZERO	4.91 1.8%	0.79 2.4%	0.03 3.7%	0.00 4.2%
	AP_SAT_SYM	4.91 1.9%	1.17 2.2%	0.03 3.7%	0.00 4.2%
	AP_WRAP	4.91 1.7%	0.04 2.3%	0.03 3.5%	0.00 4.1%
	AP_WRAP_SM	4.91 1.8%	1.18 2.2%	0.03 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	× 2.0%	1.17 2.3%	0.05 3.7%	0.00 4.2%
	AP_SAT_ZERO	× 1.8%	0.78 2.4%	0.05 3.7%	0.00 4.2%
	AP_SAT_SYM	× 2.2%	1.17 2.5%	0.05 3.7%	0.00 4.2%
	AP_WRAP	× 1.6%	0.06 2.0%	0.05 3.2%	0.00 3.7%
	AP_WRAP_SM	× 2.0%	1.18 2.4%	0.05 3.7%	0.00 4.2%
AP_RND	AP_SAT	2.87 2.0%	1.17 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.87 1.8%	0.78 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.87 2.1%	1.17 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.87 1.7%	0.02 2.3%	0.01 3.5%	0.00 4.1%
	AP_WRAP_SM	2.87 2.0%	1.18 2.3%	0.01 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	2.87 2.0%	1.17 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.87 1.8%	0.78 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.87 2.2%	1.17 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.87 1.6%	0.01 2.0%	0.01 3.2%	0.00 3.7%
	AP_WRAP_SM	2.87 2.0%	1.18 2.4%	0.01 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	2.87 1.9%	1.17 2.2%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.87 1.8%	0.78 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.87 2.1%	1.17 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.87 1.8%	0.01 2.4%	0.01 3.7%	0.00 4.2%
	AP_WRAP_SM	2.87 2.0%	1.18 2.3%	0.01 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	2.87 2.0%	1.17 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.87 1.8%	0.78 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.87 2.2%	1.17 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.87 1.6%	0.01 2.0%	0.01 3.2%	0.00 3.7%
	AP_WRAP_SM	2.87 2.0%	1.18 2.4%	0.01 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	2.87 2.0%	1.17 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.87 1.8%	0.78 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.87 2.2%	1.17 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.87 1.8%	0.01 2.4%	0.01 3.7%	0.00 4.2%
	AP_WRAP_SM	2.87 2.0%	1.18 2.4%	0.01 3.7%	0.00 4.2%

TDLC300-100 *

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	4.73 1.8%	1.03 2.1%	0.01 3.1%	0.00 3.6%
	AP_SAT_ZERO	4.73 1.8%	0.91 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	4.73 1.9%	1.03 2.2%	0.01 3.7%	0.00 4.2%
	AP_WRAP	4.73 1.7%	0.01 2.3%	0.01 3.5%	0.00 4.1%
	AP_WRAP_SM	4.73 1.8%	1.11 2.2%	0.01 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	× 2.0%	1.05 2.3%	0.08 3.7%	0.00 4.2%
	AP_SAT_ZERO	× 1.8%	0.93 2.4%	0.08 3.7%	0.00 4.2%
	AP_SAT_SYM	× 2.2%	1.05 2.5%	0.08 3.7%	0.00 4.2%
	AP_WRAP	× 1.6%	0.08 2.0%	0.08 3.2%	0.00 3.7%
	AP_WRAP_SM	× 2.0%	1.13 2.4%	0.08 3.7%	0.00 4.2%
AP_RND	AP_SAT	2.95 2.0%	1.03 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.95 1.8%	0.92 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	2.95 2.1%	1.03 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	2.95 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	2.95 2.0%	1.11 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	2.95 2.0%	1.03 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.95 1.8%	0.92 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	2.95 2.2%	1.03 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	2.95 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	2.95 2.0%	1.11 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	2.96 1.9%	1.03 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.96 1.8%	0.92 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	2.96 2.1%	1.03 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	2.96 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	2.96 2.0%	1.11 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	2.95 2.0%	1.03 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.95 1.8%	0.92 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	2.95 2.2%	1.03 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	2.95 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	2.95 2.0%	1.11 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	2.95 2.0%	1.03 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.95 1.8%	0.92 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	2.95 2.2%	1.03 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	2.95 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	2.95 2.0%	1.11 2.4%	0.00 3.7%	0.00 4.2%

TDLB100-400

Quantização	Estouro	<22,10>	<26,10>	<32,12>	<36,12>
AP_TRN	AP_SAT	3.24 1.7%	2.78 2.0%	0.00 3.3%	0.00 3.7%
	AP_SAT_ZERO	3.36 1.6%	3.00 2.2%	0.00 3.9%	0.00 4.2%
	AP_SAT_SYM	3.24 1.8%	2.78 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP	3.11 1.6%	2.40 2.1%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	4.23 1.7%	4.17 2.2%	0.00 3.9%	0.00 4.4%
AP_TRN_ZERO	AP_SAT	× 1.9%	2.88 2.3%	0.08 3.9%	0.01 4.4%
	AP_SAT_ZERO	× 1.7%	3.10 2.2%	0.08 3.9%	0.01 4.4%
	AP_SAT_SYM	× 2.1%	2.88 2.5%	0.08 3.9%	0.01 4.3%
	AP_WRAP	× 1.4%	2.52 1.9%	0.08 3.4%	0.01 3.9%
	AP_WRAP_SM	× 1.9%	4.18 2.3%	0.08 3.9%	0.01 4.4%
AP_RND	AP_SAT	3.16 1.8%	2.79 2.2%	0.01 3.9%	0.00 4.4%
	AP_SAT_ZERO	3.27 1.7%	3.01 2.2%	0.01 3.9%	0.00 4.4%
	AP_SAT_SYM	3.16 2.0%	2.79 2.4%	0.01 3.9%	0.00 4.4%
	AP_WRAP	2.98 1.6%	2.41 2.1%	0.01 3.7%	0.00 4.2%
	AP_WRAP_SM	4.23 1.9%	4.17 2.2%	0.01 3.9%	0.00 4.4%
AP_RND_ZERO	AP_SAT	3.16 1.9%	2.79 2.3%	0.01 3.9%	0.00 4.4%
	AP_SAT_ZERO	3.27 1.7%	3.01 2.2%	0.01 3.9%	0.00 4.4%
	AP_SAT_SYM	3.16 2.1%	2.79 2.5%	0.01 3.9%	0.00 4.3%
	AP_WRAP	2.98 1.4%	2.41 1.9%	0.01 3.4%	0.00 3.9%
	AP_WRAP_SM	4.23 1.9%	4.17 2.3%	0.01 3.9%	0.00 4.4%
AP_RND_MIN_INF	AP_SAT	3.16 1.8%	2.79 2.2%	0.01 3.9%	0.00 4.4%
	AP_SAT_ZERO	3.27 1.7%	3.01 2.2%	0.01 3.9%	0.00 4.2%
	AP_SAT_SYM	3.16 2.0%	2.79 2.4%	0.01 3.9%	0.00 4.3%
	AP_WRAP	2.98 1.6%	2.41 2.2%	0.01 3.9%	0.00 4.4%
	AP_WRAP_SM	4.23 1.9%	4.17 2.2%	0.01 3.9%	0.00 4.4%
AP_RND_INF	AP_SAT	3.16 1.9%	2.79 2.3%	0.01 3.9%	0.00 4.4%
	AP_SAT_ZERO	3.27 1.7%	3.01 2.2%	0.01 3.9%	0.00 4.4%
	AP_SAT_SYM	3.16 2.1%	2.79 2.5%	0.01 3.9%	0.00 4.3%
	AP_WRAP	2.98 1.4%	2.41 1.9%	0.01 3.4%	0.00 3.9%
	AP_WRAP_SM	4.23 1.9%	4.17 2.3%	0.01 3.9%	0.00 4.4%
AP_RND_CONV	AP_SAT	3.16 1.9%	2.79 2.3%	0.01 3.9%	0.00 4.4%
	AP_SAT_ZERO	3.27 1.7%	3.01 2.2%	0.01 3.9%	0.00 4.4%
	AP_SAT_SYM	3.16 2.1%	2.79 2.5%	0.01 3.9%	0.00 4.3%
	AP_WRAP	2.98 1.6%	2.41 2.2%	0.01 3.9%	0.00 4.4%
	AP_WRAP_SM	4.23 1.9%	4.17 2.3%	0.01 3.9%	0.00 4.4%